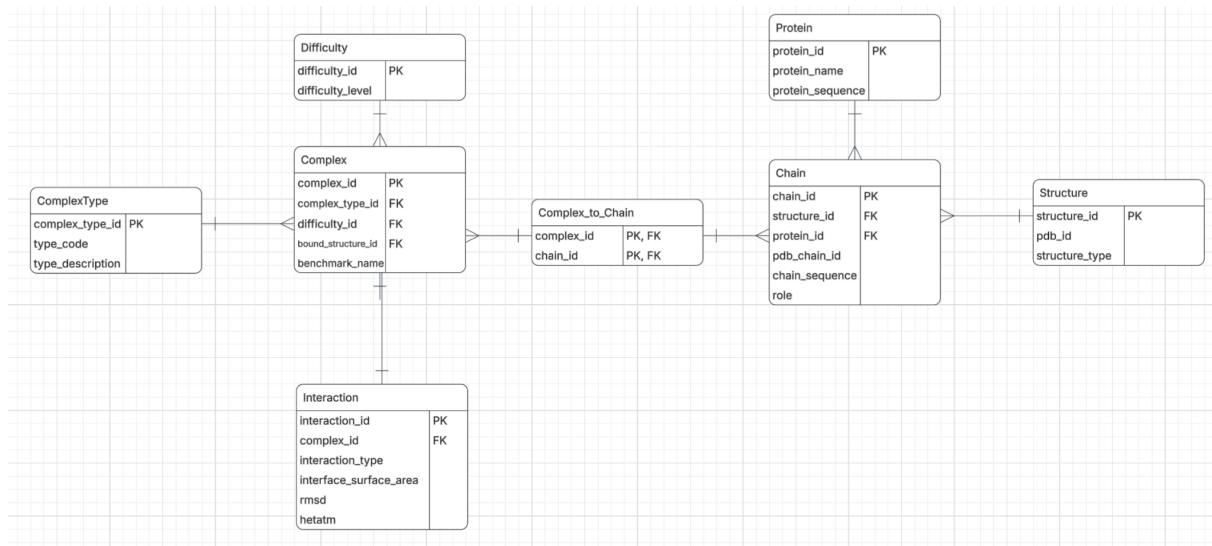
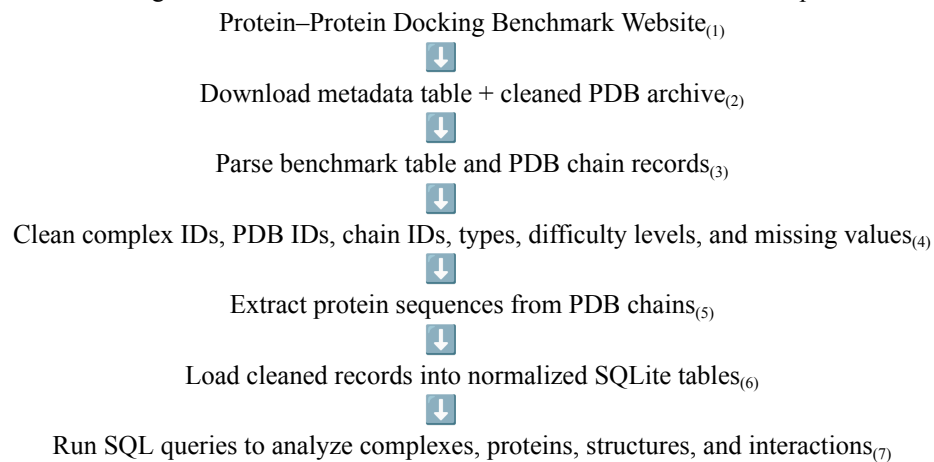


1. Conceptual Schema



2. ETL Workflow:

a. ETL Workflow Diagram *see footnotes for correlation to ETL Workflow Description*:



b. ETL Workflow Description:

- (1) Data was sourced from the Protein-Protein Docking Benchmark v5.5 website.
- (2) Benchmark metadata (Table_BM5.5.xlsx) and cleaned PDB structure files were downloaded.
- (3) Metadata was parsed to extract complex IDs, complex types, difficulty levels, and structural features (RMSD, HETATMs, interface surface area). PDB IDs and chain IDs for bound and unbound structures were also extracted.
- (4) Data was cleaned by standardizing categorical values (complex type and difficulty levels), handling missing values by storing them as NULL, and removing duplicate structures and proteins by enforcing unique IDs.
- (5) PDB files were parsed to extract protein sequences for each chain.
- (6) Cleaned data was loaded into a normalized SQLite database following foreign key dependencies:
 ComplexType → Difficulty → Structure → Protein → Chain → Complex → Complex_to_Chain → Interaction
- (7) SQL queries were executed to analyze complexes, proteins, structures, and interactions.